

Assignment 4

08022311 陈鲲龙

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Problem Description

A stream contains a waste chemical, W, with a concentration of 1 mol/liter. To meet Environmental Protection Agency and state standards, at least 90% of the chemical must be removed by reaction. The chemical decomposes by a second-order reaction with a rate constant of 1.5 liter/(mol hr). The stream flow rate is 100 liter/hour and two available reactors (400 and 2000 liters) have been placed in series (the smaller reactor is placed before the larger one).

C_{w1} = concentration in reactor 1, mol/liter

C_{w2} = concentration in reactor 2, mol/liter

F = volumetric flow rate, liter/hr

V_1 = liquid volume in reactor 1, liters

V_2 = liquid volume in reactor 2, liters

k = second-order rate constant, liter/(mol hr)

1 Mathematical Model

Consider a two-stage series CSTR with second-order reaction kinetics, where the reaction rate constant is $k = 1.5 \text{ L}/(\text{mol} \cdot \text{h})$.

Mass Balance Equations

For the two reactors, applying steady-state mass balance:

First Reactor ($V_1 = 400 \text{ L}$):

$$FC_{win} - FC_{w1} - kC_{w1}^2V_1 = 0 \quad (1)$$

Second Reactor ($V_2 = 2000 \text{ L}$):

$$FC_{w1} - FC_{w2} - kC_{w2}^2V_2 = 0 \quad (2)$$

where the flow rate is $F = 100 \text{ L/h}$, and the inlet concentration is $C_{win} = 1 \text{ mol/L}$.

2 Solving for Steady-State Concentrations

Solving the quadratic equations:

$$600C_{w1}^2 + 100C_{w1} - 100 = 0 \quad (3)$$

$$C_{w1} = \frac{-100 \pm \sqrt{100^2 + 4 \cdot 600 \cdot 100}}{2 \cdot 600} \quad (4)$$

$$= \frac{1}{3} = 0.33333 \quad (5)$$

$$3000C_{w2}^2 + 100C_{w2} - \frac{100}{3} = 0 \quad (6)$$

$$C_{w2} = \frac{-100 \pm \sqrt{100^2 + 4 \cdot 3000 \cdot \frac{100}{3}}}{2 \cdot 3000} \quad (7)$$

$$= 0.09005 \quad (8)$$

Removal efficiency:

$$1 - \frac{C_{w2}}{C_{win}} = 90.995\% > 90\% \quad (9)$$

3 State-Space Model

Defining deviation variables:

$$x = \begin{bmatrix} C_{w1} - C_{w1s} \\ C_{w2} - C_{w2s} \end{bmatrix}, \quad u = \begin{bmatrix} F - F_s \\ C_{win} - C_{wins} \end{bmatrix}$$

Nonlinear State Equations:

The dynamic mass balance equations for the two reactors are given as:

First Reactor:

$$\frac{dC_{w1}}{dt} = \frac{F}{V_1}(C_{win} - C_{w1}) - kC_{w1}^2 \quad (10)$$

Second Reactor:

$$\frac{dC_{w2}}{dt} = \frac{F}{V_2}(C_{w1} - C_{w2}) - kC_{w2}^2 \quad (11)$$

Linearized state equation:

$$\dot{x} = Ax + Bu \quad (12)$$

where:

$$A = \begin{bmatrix} \frac{\partial}{\partial C_{w1}} \left(\frac{dC_{w1}}{dt} \right) & \frac{\partial}{\partial C_{w2}} \left(\frac{dC_{w1}}{dt} \right) \\ \frac{\partial}{\partial C_{w1}} \left(\frac{dC_{w2}}{dt} \right) & \frac{\partial}{\partial C_{w2}} \left(\frac{dC_{w2}}{dt} \right) \end{bmatrix} = \begin{bmatrix} -\frac{F}{V_1} - 2kC_{w1} & 0 \\ \frac{F}{V_2} & -\frac{F}{V_2} - 2kC_{w2} \end{bmatrix} \quad (13)$$

$$B = \begin{bmatrix} \frac{\partial}{\partial F} \left(\frac{dC_{w1}}{dt} \right) & \frac{\partial}{\partial C_{win}} \left(\frac{dC_{w1}}{dt} \right) \\ \frac{\partial}{\partial F} \left(\frac{dC_{w2}}{dt} \right) & \frac{\partial}{\partial C_{win}} \left(\frac{dC_{w2}}{dt} \right) \end{bmatrix} = \begin{bmatrix} \frac{(C_{win} - C_{w1})}{V_1} & \frac{F}{V_1} \\ \frac{(C_{w1} - C_{w2})}{V_2} & 0 \end{bmatrix} \quad (14)$$

4 Show A & B Matrices

Substituting the values of each variable into equations (13) and (14), we obtain:

$$A = \begin{bmatrix} -1.25 & 0 \\ 0.05 & -0.32015 \end{bmatrix}, \quad B = \begin{bmatrix} 0.001667 & 0.25 \\ 0.0001216 & 0 \end{bmatrix} \quad (15)$$

5 Eigenvalue Analysis

Find the eigenvalues of A using the MATLAB eig function:

MATLAB code:

$$A = [-1.25 \ 0; 0.05 \ -0.32015]; \quad \text{eig}(A) \quad (16)$$

result:

```
>> A = [-1.25 0; 0.05 -0.32015]; eig(A)
```

```
ans =
```

```
-0.3201
```

```
-1.2500
```

Find the eigenvalues by hand, by solving $\det(1 - A) = 0$:

Solving $\det(\lambda I - A) = 0$:

$$(\lambda + 1.25)(\lambda + 0.32015) = 0 \quad (17)$$

$$\lambda_1 = -1.25, \quad \lambda_2 = -0.32015 \quad (18)$$

Since $\lambda_1, \lambda_2 < 0$, the system is asymptotically stable.

6 Linear vs Nonlinear

Solve 1 for the nonlinear differential equations, using ode45. Compare the linear and nonlinear variables on the same plots (make certain you convert from deviation to physical variables for the linear results) for a step change in the flow rate from 100 liters/hour to 110 liters/hour. Assume the initial concentrations are the steady-state values (0.3333 and 0.09005). Compare the linear and nonlinear responses of the reactor concentrations. Is the removal specification still obtained?

MATLAB code:

```
clc; clear; close all;
```

```
% Constants
```

```
F0 = 100; % Initial flow rate (L/h)
```

```
F_new = 110; % New flow rate after step change (L/h)
```

```
V1 = 400; % Volume of reactor 1 (L)
```

```
V2 = 2000; % Volume of reactor 2 (L)
```

```
k = 1.5; % Reaction rate constant (L/(mol · h))
```

```
% Steady-state concentrations
```

```
Cw1s = 0.3333; % Reactor 1 steady-state concentration (mol/L)
```

```
Cw2s = 0.09005; % Reactor 2 steady-state concentration (mol/L)
```

```
Cwin = 1; % Inlet concentration (mol/L)
```

```

% Time span
tspan = [0 50]; % Simulate for 50 hours

% Initial conditions (steady-state concentrations)
C0 = [Cw1s; Cw2s];

% Solve nonlinear system using ode45
[t_nl, C_nl] = ode45(@(t, C) nonlinear_reactor(t, C, F_new, V1, V2, k, Cwin), tspan, C0);

% Compute A and B matrices at steady-state
% A = [-F0/V1 - 2*k*Cw1s,  F0/V1;
%      F0/V2, -F0/V2 - 2*k*Cw2s];
%
% B = [1/V1, 0;
%      0,    0];
A = [-F0/V1 - 2*k*Cw1s,  0;
      F0/V2, -F0/V2 - 2*k*Cw2s];

B = [(Cwin-Cw1s)/V1, F0/V1;
      (Cw1s-Cw2s)/V2,  0];

% Solve linearized system using ode45
x0 = [0; 0]; % Deviation variables start at zero
[t_lin, x_lin] = ode45(@(t, x) A * x + B * [(F_new - F0); 0], tspan, x0);

% Convert linearized deviation variables back to physical concentrations
C_lin = x_lin + [Cw1s, Cw2s];

% Plot results
figure;
subplot(2,1,1);
plot(t_nl, C_nl(:,1), 'b', 'LineWidth', 1.5); hold on;
plot(t_lin, C_lin(:,1), 'r--', 'LineWidth', 1.5);
xlabel('Time (h)');
ylabel('C_{w1} (mol/L)');
legend('Nonlinear', 'Linearized');
title('Reactor 1 Concentration Response');

subplot(2,1,2);
plot(t_nl, C_nl(:,2), 'b', 'LineWidth', 1.5); hold on;
plot(t_lin, C_lin(:,2), 'r--', 'LineWidth', 1.5);
xlabel('Time (h)');

```

```

ylabel('C_{w2} (mol/L)');
legend('Nonlinear', 'Linearized');
title('Reactor 2 Concentration Response');

grid on;

% Check if removal specification is still met
final_Cw2_nl = C_nl(end,2);
removal_nl = (1 - final_Cw2_nl) * 100; % Percentage removal
disp(['Final nonlinear Cw2: ', num2str(final_Cw2_nl)]);
disp(['Nonlinear removal efficiency: ', num2str(removal_nl), '%']);

% Function for nonlinear reactor ODEs
function dCdt = nonlinear_reactor(~, C, F, V1, V2, k, Cwin)
    Cw1 = C(1);
    Cw2 = C(2);

    dCw1dt = (F/V1) * (Cwin - Cw1) - k * Cw1^2;
    dCw2dt = (F/V2) * (Cw1 - Cw2) - k * Cw2^2;

    dCdt = [dCw1dt; dCw2dt];
end

```

Result:

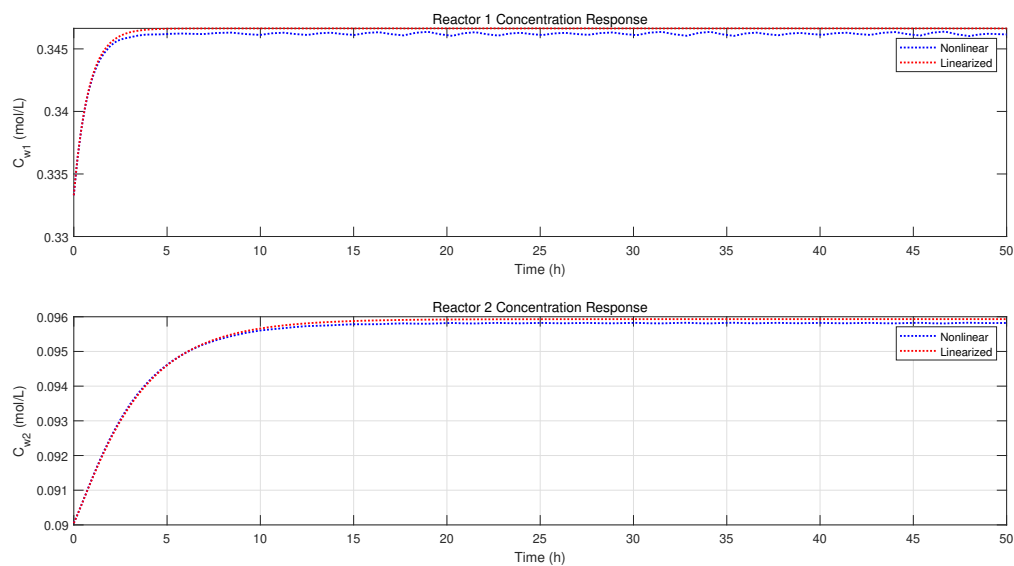


Fig. 1: Linear vs Nonlinear

Removal efficiency:

$$1 - \frac{C_{w2}}{C_{win}} \approx 1 - \frac{0.096}{1} = 90.4\% > 90\% \quad (19)$$

Conclusion:

Therefore, the removal specification is still obtained.

Comparison:

A linear state-space model approximates the behavior of the reactor by assuming that the system operates near a steady-state operating point, where deviations are small. The linearization process involves linearizing the system around the steady-state values of concentrations and flow rate.

However, in a nonlinear model, the differential equations describe the actual dynamic behavior of the reactor without simplifying assumptions. The concentrations and flow rate are related by a set of nonlinear equations that account for real chemical reaction kinetics and mass transfer limitations.

The key difference between the linear and nonlinear models is that the linear model assumes small deviations from the steady-state, and thus does not capture the nonlinearities in the system at larger deviations. This could lead to discrepancies between the linear and nonlinear concentration responses, especially when the system is subjected to larger flow rate changes or when operating outside of the linear region.

By comparing the results, we can assess whether the linear approximation still provides an accurate representation of the system behavior, or if the nonlinear dynamics must be considered for precise control and accurate concentration predictions.

From my perspective, the linear state-space model provides a more complete description of the system, as it takes into account the initial state, namely, the system's zero-input response. In contrast, differential equations and their transfer functions assume an initial state of zero, only describing the system's zero-state response.

7 Reverse the order of the reaction vessels

Would better steady-state removal of W be obtained if the order of the reaction vessels was reversed? Why or why not? (Show your calculations.)

The dynamic mass balance equations for the two reactors will turn into:

First Reactor:

$$\frac{dC'_{w1}}{dt} = \frac{F}{V_2}(C_{win} - C'_{w1}) - kC'^2_{w1} \quad (20)$$

Second Reactor:

$$\frac{dC'_{w2}}{dt} = \frac{F}{V_1}(C'_{w1} - C'_{w2}) - kC'^2_{w2} \quad (21)$$

Solving the quadratic equations:

$$3000C'^2_{w1} + 100C'_{w1} - 100 = 0 \quad (22)$$

$$C'_{w1} = \frac{-100 \pm \sqrt{100^2 + 4 \cdot 3000 \cdot 100}}{2 \cdot 3000} \quad (23)$$

$$= \frac{1}{6} = 0.16667 \quad (24)$$

$$600C'_{w2} + 100C'_{w2} - \frac{100}{6} = 0 \quad (25)$$

$$C'_{w2} = \frac{-100 \pm \sqrt{100^2 + 4 \cdot 600 \cdot \frac{100}{6}}}{2 \cdot 600} \quad (26)$$

$$= 0.103 \quad (27)$$

Removal efficiency:

$$1 - \frac{C_{w2}}{C_{win}} = 89.7\% < 90\% \quad (28)$$

Result:

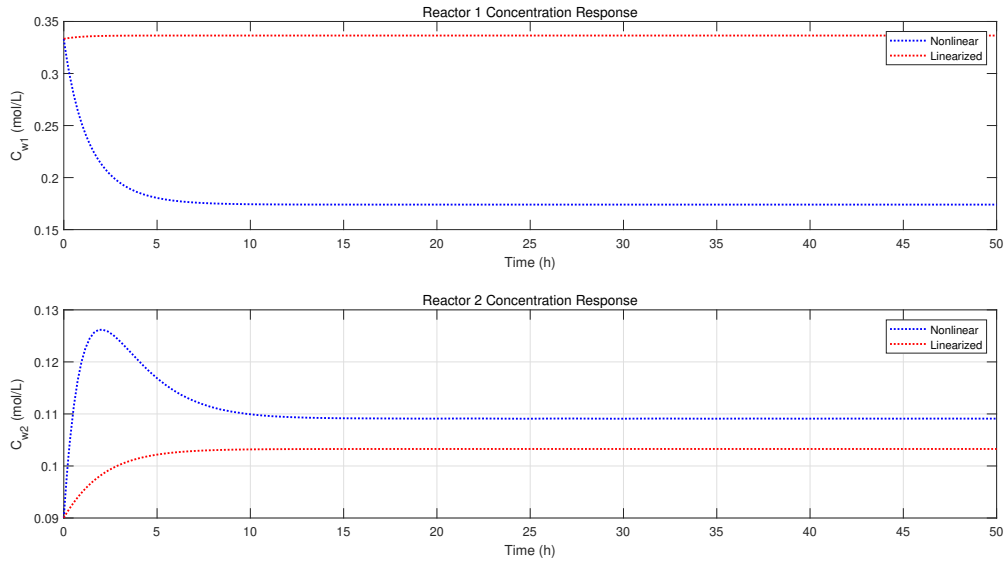


Fig. 2: Reverse the order of the reaction vessels

Conclusion:

Reversing the order of the reaction vessels would **not** result in better steady-state removal of W. In fact, the outcome after swapping the vessels would **fail to meet** the required specification.